

Regime-Dependent Basis Attribution in Gold Futures: Why Sign-Flipping is Not Noise*

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Abstract

Rolling-window factor models for the gold futures basis exhibit persistent coefficient sign-flipping: the DXY dollar index coefficient is positive in some periods and negative in others, with only 64% sign consistency over a decade. We show this instability is regime-dependent, not noise. Using a dual Hidden Markov Model framework—one classifying market risk conditions (VIX and DXY levels) and another classifying CFTC positioning regimes (commercial-led versus speculator-led)—we identify market states in which factor signs become statistically consistent. In the Risk-Off/Commercial-Led regime, VIX sign consistency rises from 59% to 75% (bootstrap 95% CI excludes coin-toss; overlap-adjusted z -test $p = 0.046$). Critically, F -tests show no evidence that regimes alter mean coefficient magnitudes ($p = 0.531$), while Brown-Forsythe tests confirm that regimes significantly alter coefficient *dispersion* ($p < 0.001$ for both DXY and VIX). The regime effect operates on the temporal distribution of coefficient signs, not on coefficient levels—a second-moment effect that we directly measure, not merely infer. This framework adds interpretive value, informing practitioners *when* to trust factor attribution, without improving point forecasts.

Keywords: gold futures, basis attribution, regime switching, Hidden Markov Model, sign consistency, rolling regression, CFTC positioning

JEL Codes: G13, G14, C22, C58

Data Availability: Data and code available from the author upon request.

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1 Introduction

Factor attribution models are a cornerstone of commodity market analysis. Practitioners use rolling regressions to decompose daily price movements into contributions from macroeconomic drivers—the dollar, volatility, inflation expectations, and others. For gold futures, such models typically explain 30–40% of daily basis variation, a useful level for generating narrative intelligence about why the spread between futures and spot prices moved on a given day.

However, these models suffer from a persistent practical problem: *coefficient sign-flipping*. The DXY dollar index—one of the most economically important drivers of gold—carries a positive coefficient in some rolling windows and a negative coefficient in others. Over a decade of daily data (2015–2025), the DXY coefficient has a consistent sign in only 64% of 30-day rolling windows. The VIX volatility index is worse, at 59%. For a practitioner interpreting today’s attribution, this means that on roughly one-third to two-fifths of all days, the model’s stated direction of a factor’s influence may be unreliable.

The standard response to sign-flipping is to treat it as noise and either (i) remove the offending variables, (ii) increase the window length to smooth out instability, or (iii) report attribution without directional confidence qualifications. We show that all three responses are suboptimal.

1.1 Contribution

This paper makes three contributions.

First, we demonstrate that coefficient sign-flipping in rolling-window basis attribution is *regime-dependent*, not random. A dual Hidden Markov Model (HMM) framework—combining a market risk regime (fitted on VIX and DXY levels) with a CFTC positioning regime (fitted on speculator dominance and managed money concentration)—identifies four market states. Within each state, sign consistency improves substantially. VIX sign consistency rises from 59% (pooled) to 75% in the Risk-Off/Commercial-Led regime, a 16-percentage-point improvement. After correcting for the serial dependence inherent in overlapping 30-day windows (effective $N \approx 90$, versus 2,004 raw windows), this improvement remains significant at the 5% level ($z_{\text{adj}} = 1.684$, $p = 0.046$). A permutation test over 1,000 random label shuffles confirms $p < 0.001$.

Second, we show that the regime effect operates at the *temporal* level, not the *cross-sectional* level. Joint F -tests for coefficient equality across regime cells are not significant ($p = 0.531$): the mean coefficient vectors do not differ across regimes. However, Brown-Forsythe tests show that the *variance* of rolling coefficients differs highly significantly across regimes ($p < 0.001$ for both DXY and VIX). What differs is the *sign distribution*—regimes alter the coefficient distribution relative to zero, narrowing it in risk-off conditions and widening it in calm markets, thereby shifting probability mass toward or away from

one side. This is a second-moment effect (dispersion change), not a first-moment effect (mean shift), and it is precisely what standard structural break tests are not designed to detect.

Third, we establish that the regime framework adds *interpretive* value without *predictive* value. The regime-conditional model produces identical point predictions to the pooled model (Diebold-Mariano statistic = 0.000, $p = 1.0$). Its value lies in telling practitioners *when* a factor’s attributed direction is reliable—Risk-Off/Commercial-Led conditions produce DXY and VIX sign consistency above 70%, while Risk-On/Commercial-Led conditions leave VIX near coin-toss (50%).

1.2 Summary of Findings

Using 2,543 trading days of gold futures data (January 2015 to December 2025) and 2,004 rolling 30-day OLS windows, we find:

- Removing sign-flipping variables reduces model R^2 by approximately two-thirds (from 34.7% to 11.6%), confirming they carry real information despite directional instability (Section 4).
- CFTC positioning data adds 5.1 pp to R^2 as a direct regressor but has coin-toss sign consistency (50–56%), making it suitable as a regime indicator rather than a factor (Section 4).
- Bootstrap 95% confidence intervals for within-regime sign consistency exclude 0.50 (coin-toss) in all four regime cells for both DXY and VIX, though margins vary: the weakest cell (Risk-On/Commercial-Led VIX) barely excludes 0.50 with a lower bound of 50.3% (Section 6).
- Brown-Forsythe tests confirm that the *variance* of rolling coefficients differs significantly across regime cells ($p < 0.001$ for both DXY and VIX), providing direct evidence for the second-moment mechanism (Section 6.5).
- Overlap-adjusted two-proportion z -tests (effective $N \approx 90$ for VIX) confirm that VIX sign consistency in Risk-Off/Commercial-Led exceeds the pooled level ($p = 0.046$). Individual DXY cell comparisons do not survive the overlap correction (Section 6).
- A permutation test (1,000 label shuffles) confirms that HMM labels produce significantly higher sign consistency than random partitions ($p < 0.001$ overall; VIX best cell 7.5σ above null) (Section 6.7).
- Out-of-sample validation across two train/test splits shows the regime framework generalizes: the headline Risk-Off/Commercial-Led cell maintains stable R^2 (38.8% train vs. 36.6% test) and sign consistency patterns persist (Section 7).
- The Durbin-Watson statistic of 2.88 indicates moderate negative serial correlation in full-sample OLS residuals, justifying the rolling-window approach (Section 7).

1.3 Paper Outline

Section 2 reviews related literature on commodity basis models, regime-switching in finance, CFTC positioning, and coefficient instability. Section 3 describes the data, rolling OLS specification, sign consistency metric, and dual-HMM design. Section 4 presents motivating evidence from variable pruning and COT screening experiments. Section 5 describes the regime classification. Section 6 presents the core results on regime-dependent sign consistency, provides direct evidence that the regime effect operates on coefficient dispersion (Brown-Forsythe tests), reconciles the paradox between significant sign consistency improvements and non-significant F -tests, reports overlap-corrected inference for two-proportion comparisons, addresses multiple testing, and validates HMM label quality via a permutation test. Section 7 provides robustness checks including out-of-sample validation, benchmark comparisons, a VIX-threshold comparison, and an event study. Section 8 discusses economic interpretation, practical implications, and limitations. Section 9 concludes.

2 Related Literature

This paper connects to four strands of the literature: commodity basis and convenience yield theory, regime-switching models in finance, the role of CFTC positioning data, and coefficient instability in rolling regressions.

2.1 Commodity Basis and Convenience Yield

The gold futures basis—the difference between the futures price and the spot price—has been studied through the lens of storage theory since [Working \(1949\)](#) and [Brennan \(1958\)](#). The basis reflects carrying costs (financing, storage, insurance) minus the convenience yield of holding the physical commodity. [Fama and French \(1987\)](#) showed that commodity futures prices contain information about expected spot prices and risk premiums, while [Gorton et al. \(2013\)](#) demonstrated that the basis (or “roll return”) is the primary driver of commodity futures returns. [Erb and Harvey \(2006\)](#) established that commodity futures offer strategic diversification value but that returns are driven by basis dynamics rather than spot price appreciation.

For gold specifically, [O’Connor et al. \(2015\)](#) provides a comprehensive survey of the financial economics of gold, noting that gold’s unique dual nature—monetary asset and physical commodity—makes its basis dynamics distinct from agricultural or energy commodities. Our contribution extends this literature by showing that the macro factor model for gold basis variation has regime-dependent coefficient signs, a phenomenon not previously documented.

2.2 Regime-Switching Models in Finance

Hamilton (1989) introduced Markov-switching models to economics, demonstrating that business cycle dynamics are well captured by discrete regime changes. This framework was extended to interest rates by Ang and Bekaert (2002) and to commodity markets by Alizadeh et al. (2008), who applied regime switching to energy hedging, and Chevallier (2011), who detected regime instability in carbon markets.

For gold, Baur and Lucey (2010) and Baur and McDermott (2010) showed that gold's role as a safe haven is regime-dependent—gold is a hedge during normal times but a safe haven during extreme market stress. Reboredo (2013) extended this to the gold-dollar relationship, finding that gold's hedging properties against the dollar depend on market conditions.

Our dual-HMM framework departs from this literature in two ways. First, we use *two independent* HMMs (market risk and positioning) rather than a single regime indicator, creating a 2×2 interaction matrix. Second, we use HMMs for *interpretive* rather than predictive purposes—identifying when factor attribution signs are reliable, not generating trading signals.

2.3 CFTC Positioning and Market Microstructure

The Commodity Futures Trading Commission's Commitments of Traders (COT) report has been widely studied as a source of information about market positioning. Büyüksahin and Harris (2011) examined whether speculators drive crude oil prices, while Brunetti et al. (2016) studied the relationship between speculative activity and price volatility. De Roon et al. (2000) established that hedging pressure effects are significant in futures markets, and Kang et al. (2020) decomposed commodity futures premiums into hedger and speculator components.

Our finding that COT positioning data performs poorly as a direct regression variable (sign consistency 50–56%) but effectively identifies the positioning regime via HMM contributes to this literature by suggesting that COT data's value is structural (identifying *who* dominates the market) rather than directional (predicting *which way* the market moves).

2.4 Coefficient Instability and Rolling Regressions

The instability of regression coefficients in financial time series is well documented. Pesaran and Timmermann (2002) showed that return prediction under model instability requires adaptive methods, while Rossi (2013) surveyed advances in forecasting under structural change. Andrews (1993) and Bai and Perron (1998) developed formal tests for parameter instability and multiple structural breaks.

Our sign consistency metric is related to but distinct from these approaches. Traditional structural break tests (Chow, Bai-Perron, Andrews) test whether coefficient *levels* change at specific or unknown breakpoints. Sign consistency measures whether the coefficient’s *direction* (above or below zero) is stable across rolling windows. As we demonstrate, these can diverge: coefficient levels may be statistically indistinguishable across regimes (F -test $p = 0.531$) while coefficient *dispersion* differs highly significantly (Brown-Forsythe $p < 0.001$). This distinction—between first-moment and second-moment regime effects—is, to our knowledge, novel.

3 Data and Methodology

3.1 Data Description

Our sample spans January 5, 2015 to December 31, 2025—2,543 trading days. Table 1 describes the variables.

Table 1: Data Description

Variable	Source	Transform	Role
Gold futures (GC)	Databento/Yahoo	Level	Futures price
GLD ETF $\times 10.6759$	Yahoo Finance	Level	Spot proxy ^a
DXY (Dollar Index)	Investing.com CSV	% change	Macro factor
VIX	Yahoo Finance	Absolute Δ	Macro factor
Breakeven inflation (T10YIE)	FRED	$\Delta \times 100$	Macro factor
USD/CNY	Yahoo Finance	% change	Demand proxy
USD/INR	Yahoo Finance	% change	Demand proxy
GLD Volume	Yahoo Finance	Log change	Flow proxy
GVZ (Gold Vol Index)	FRED (GVZCLS)	Absolute Δ	Sentiment
Volume change	Yahoo (GC=F vol.)	% change	Flow proxy
Spec dominance ratio ^b	CFTC COT	Level	HMM feature
MM concentration ^b	CFTC COT	Level	HMM feature

^a GLD ETF closing price multiplied by a calibrated constant (10.6759) to approximate London gold spot. The multiplier is estimated via OLS regression of CME settlement prices on GLD prices over the full sample.

^b CFTC Disaggregated Futures-Only report, weekly (as-of Tuesday, published Friday), forward-filled to daily frequency. 1,000 weeks of history (2007–2026). Spec dominance ratio = $|\text{net speculative}| / (|\text{net speculative}| + |\text{net commercial}|)$. MM concentration = net managed money / total open interest.

Data quality note. The majority of market data (VIX, ETF prices, currency pairs) is sourced from Yahoo Finance, which provides free 15-minute delayed data. While institutional data providers (Bloomberg, Refinitiv) offer higher quality feeds, Yahoo Finance was chosen for cost-effectiveness and reproducibility—any researcher can replicate our data pipeline without a commercial license. No interpolation, imputation, or data massaging

is applied to the Yahoo Finance data; observations with missing values are dropped via inner join across all series. The DXY historical series is sourced from Investing.com CSV downloads. FRED provides breakeven inflation and GVZ data at daily frequency with no known quality issues.

Table 2 reports summary statistics for the dependent variable and regressors.

Table 2: Summary Statistics (2,543 Trading Days)

Variable	Mean	SD	Min	Median	Max
Basis change (\$)	0.09	12.86	-151.63	-0.03	149.61
DXY change (%)	0.01	0.45	-2.15	0.01	2.65
VIX change (pts)	0.00	2.00	-18.71	-0.10	24.86
Breakeven $\Delta \times 100$	-0.01	3.10	-32.00	0.00	25.00
USD/CNY change (%)	0.01	0.30	-2.14	0.00	2.14
USD/INR change (%)	0.02	0.37	-2.29	0.00	2.98
GLD vol change (log)	0.00	0.43	-2.63	-0.02	3.19
GVZ change (pts)	0.00	0.95	-9.50	-0.07	7.25

Volume change excluded due to extreme outliers from percentage-change calculation on near-zero denominators.

3.2 Basis Construction

We define the gold futures basis as:

$$\text{basis}_t = P_t^{\text{futures}} - P_t^{\text{spot}} \quad (1)$$

where $P_t^{\text{spot}} = \text{GLD}_t \times 10.6759$. The futures price is constructed from an unadjusted continuous front-month GC contract series (Yahoo Finance ticker GC=F), which rolls to the next active contract approximately 3–5 days before first-notice day.¹ The dependent variable in all regressions is the daily basis change, $\Delta \text{basis}_t = \text{basis}_t - \text{basis}_{t-1}$. Working with changes rather than levels addresses non-stationarity in the basis series.

3.3 Rolling OLS Attribution Model

For each trading day t , we estimate an OLS regression over the preceding $w = 30$ trading days:

$$\Delta \text{basis}_s = \alpha + \sum_{k=1}^K \beta_k x_{k,s} + \varepsilon_s, \quad s \in [t - w + 1, t] \quad (2)$$

¹The extreme Δbasis observations in Table 2 (± 150 range) correspond to contract-roll dates when the front-month definition jumps between expiries. These roll artifacts are retained without winsorization; a post-hoc check confirms they do not cluster disproportionately in any single regime cell, and their inclusion does not materially alter the headline sign-consistency results.

where $K = 8$ features (DXY change, VIX change, breakeven change, USD/CNY change, USD/INR change, ETF volume change, GVZ change, volume change). This yields 2,004 rolling windows over the 2,543-day sample.

The 30-day window balances responsiveness to regime shifts against estimation noise. The Durbin-Watson statistic of 2.88 on full-sample OLS residuals (Section 7) indicates moderate negative serial correlation, suggesting that shorter windows better capture the time-varying structure than an expanding window or static model.

3.4 Sign Consistency

We define sign consistency for feature k across a set of N rolling windows as:

$$SC_k = \max\left(\frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{\beta}_{k,i} > 0], \frac{1}{N} \sum_{i=1}^N \mathbf{1}[\hat{\beta}_{k,i} < 0]\right) \quad (3)$$

where $\hat{\beta}_{k,i}$ is the estimated coefficient for feature k in window i . Sign consistency ranges from 0.50 (exactly half positive, half negative—a coin-toss) to 1.00 (all coefficients have the same sign).

This metric captures *directional reliability*: when SC_k is high, a practitioner can state with confidence that “factor k pushed the basis in direction d today,” because the model consistently assigns that direction to factor k in the current market conditions.

3.5 Dual-HMM Regime Framework

We fit two independent Gaussian Hidden Markov Models, each with two states.

Market Risk HMM. Features: VIX level and DXY level (daily frequency). The two states are labeled Risk-On (low VIX ≈ 13.9 , DXY ≈ 98.9 ; 1,342 days, 52.8% of sample) and Risk-Off (high VIX ≈ 23.7 , DXY ≈ 97.9 ; 1,201 days, 47.2%). The transition matrix exhibits high persistence: $P(\text{Risk-Off} \rightarrow \text{Risk-Off}) = 0.983$, $P(\text{Risk-On} \rightarrow \text{Risk-On}) = 0.986$.

Positioning HMM. Features: speculator dominance ratio and managed money concentration (weekly CFTC data, forward-filled to daily). The two states are Commercial-Led (spec dominance ≈ 0.46 , MM concentration ≈ 0.27 ; 800 weeks, 80%) and Speculator-Led (spec dominance ≈ 0.49 , MM concentration ≈ 0.05 ; 200 weeks, 20%). Commercial-Led is the dominant state; Speculator-Led episodes are less frequent but persistent once initiated ($P(\text{Spec} \rightarrow \text{Spec}) = 0.950$).

Interaction matrix. Crossing the two binary regimes produces four cells: Risk-On/Commercial-Led (806 windows), Risk-On/Speculator-Led (209), Risk-Off/Commercial-Led (784), and

Risk-Off/Speculator-Led (205). Each rolling window is assigned to a cell based on the regime label of its final day.

Both HMMs use `GaussianHMM` from `hmmlearn` with full covariance matrices, `StandardScaler` preprocessing, 200 iterations, and `random_state=42` for reproducibility.

3.6 Statistical Testing Framework

We employ seven classes of tests to validate the regime framework:

1. **Block bootstrap** (Künsch, 1989; Efron and Tibshirani, 1993): 1,000 resamples with block size 30 (matching the rolling window) to construct 95% confidence intervals for within-regime sign consistency and mean coefficients. The block structure properly accounts for the serial dependence induced by overlapping windows.
2. **F-test**: Restricted (pooled OLS, $K + 1$ parameters) versus unrestricted (separate OLS per regime cell, $4(K + 1)$ parameters). Tests whether regime conditioning improves model fit in a full-sample static framework.
3. **Chow test** (Chow, 1960): Pairwise structural break tests between all six pairs of regime cells.
4. **Two-proportion z-test** (overlap-corrected): Tests whether within-regime sign consistency exceeds pooled sign consistency. Because consecutive 30-day rolling windows share 29 of 30 observations, the 2,004 raw windows are not independent. We compute effective sample sizes via autocorrelation of the sign indicator series (effective $N \approx 90\text{--}120$, an approximately $17\text{--}22\times$ reduction).
5. **Diebold-Mariano test** (Diebold and Mariano, 1995): Compares predictive accuracy between model pairs using Newey-West HAC standard errors (Newey and West, 1987).
6. **Permutation test**: Shuffles regime cell labels across rolling windows 1,000 times, comparing real sign consistency to the null distribution under random labeling. This test has built-in multiplicity control for the best-cell statistic.
7. **Brown-Forsythe test** (Brown and Forsythe, 1974): Tests whether the *variance* of rolling coefficients differs across regime cells. This directly tests the second-moment hypothesis: if regimes narrow the coefficient distribution relative to zero, coefficient dispersion should differ across cells.

4 Motivating Evidence

Before introducing the regime framework, we present two experiments that motivate its necessity.

4.1 The Sign-Flipping Problem

Table 3 reports sign consistency for all eight features in the pooled model (no regime conditioning).

Table 3: Pooled Model: Sign Consistency and Mean Coefficients

Feature	Sign Consistency	Dominant Sign	Mean Coefficient
DXY change	64.2%	+	+1.435
Volume change	58.5%	+	+0.001
VIX change	58.6%	−	−0.302
USD/INR change	58.1%	−	−2.817
USD/CNY change	56.1%	+	+1.873
GVZ change	56.0%	−	−1.297
Breakeven change	55.4%	+	+0.224
GLD volume change	50.9%	+	+0.676

No feature exceeds 65% sign consistency. GLD volume change is at the coin-toss level (50.9%). For practical attribution, this means that on any given day, the model’s stated direction of a factor’s influence has a 35–50% chance of being the opposite sign from the typical direction.

4.2 Why Naïve Pruning Fails

A natural response is to remove the sign-flipping variables and keep only those with strong directional stability. We tested this hypothesis by removing the four weakest variables (DXY, VIX, USD/CNY, GLD volume) and retaining only breakeven inflation, USD/INR, and volume change.

Table 4: Effect of Variable Pruning on Model Performance

Metric	Full Model (7 var) ^a	Pruned Model (3 var)
Mean R^2	30.7%	11.6%
Median R^2	29.9%	9.6%
Max VIF	1.10	1.00

^aA 7-variable illustration (GVZ excluded); the canonical 8-variable model achieves 34.7% mean R^2 . Pruning reduces R^2 by approximately two-thirds (34.7% $\rightarrow$ 11.6%), with the 8-variable result directionally identical.

Removing sign-flipping variables reduces explanatory power by approximately two-thirds (from 34.7% in the canonical 8-variable model to 11.6% with only 3 variables). All pairwise VIF values are ≤ 1.10 , confirming the removed variables carry independent information—they are not redundant with the retained features. The sign-flipping vari-

ables are valuable; the problem is that their directional interpretation changes across market conditions.

4.3 COT Positioning as Regime Indicator, Not Factor

Given that the sign-flipping variables carry real information, we next tested whether CFTC Commitments of Traders (COT) data could improve the model. Adding both speculator dominance change and managed money concentration change as direct regressors increases R^2 by 5.1 pp (from 34.7% to 39.8%). However, these COT variables themselves have coin-toss sign consistency (50–56%).

This pattern—substantial R^2 contribution with poor sign consistency—matches exactly the behavior of DXY and VIX. The COT data identifies the market *regime* (commercial-led versus speculator-led), which in turn determines how other variables affect the basis. It is a regime indicator, not a direct factor. This insight motivates the positioning HMM in Section 5.

5 Regime Classification

5.1 Market Risk Regime

Table 5 reports the Market Risk HMM parameters.

Table 5: Market Risk HMM: Emission Parameters and Transition Matrix

State	VIX Mean	DXY Mean	Days	% of Sample
Risk-On	13.9	98.9	1,342	52.8%
Risk-Off	23.7	97.9	1,201	47.2%
<i>Transition matrix:</i>				
	→ Risk-Off	→ Risk-On		
Risk-Off	0.983	0.017		
Risk-On	0.014	0.986		

The two states are well separated on VIX (9.8-point gap) with a modest DXY differential. Both states are highly persistent: the expected duration is $1/(1 - 0.983) \approx 59$ days for Risk-Off and $1/(1 - 0.986) \approx 71$ days for Risk-On. This persistence means regime transitions are infrequent but meaningful when they occur.

5.2 Positioning Regime

Table 6 reports the Positioning HMM parameters.

Commercial-Led is the dominant state (80% of the sample). The state labels merit discussion: the speculator dominance ratio differs only modestly across states (0.463 vs.

Table 6: Positioning HMM: Emission Parameters and Transition Matrix

State	Spec Dominance	MM Concentration	Weeks	% of Sample
Commercial-Led	0.463	0.268	800	80.0%
Speculator-Led	0.486	0.049	200	20.0%
<i>Transition matrix:</i>				
	→ Comm-Led	→ Spec-Led		
Comm-Led	0.989	0.011		
Spec-Led	0.050	0.950		

0.486), so the label “Speculator-Led” should not be read as speculator dominance in absolute terms. What primarily separates the states is *managed money concentration* (0.268 vs. 0.049—a nearly $5\times$ difference). In the Commercial-Led state, managed money holds concentrated net long positions alongside commercial hedging activity. In the Speculator-Led state, managed money positioning is dispersed and closer to flat. We retain the “Commercial-Led” / “Speculator-Led” labels for readability, but the operative distinction is MM concentration, not speculator share.

5.3 Interaction Matrix

Table 7 shows the R^2 and window counts for each regime cell.

Table 7: Interaction Matrix: R^2 by Regime Cell

	Commercial-Led		Speculator-Led	
	R^2	Windows	R^2	Windows
Risk-Off	36.7%	784	29.0%	205
Risk-On	34.7%	806	32.8%	209
Pooled	34.7%	2,004		

Risk-Off/Commercial-Led has the highest R^2 (36.7%)—hedging flows in stressed markets are the most predictable by macro variables. Risk-Off/Speculator-Led has the lowest (29.0%)—speculator-driven panic is harder to explain with standard macro factors. The cell sizes range from 205 to 806 windows, providing reasonable statistical power for all cells except Risk-Off/Speculator-Led, where results should be interpreted with caution.

6 Core Results: Sign Consistency is Regime-Dependent

6.1 Within-Regime Sign Consistency

Table 8 presents sign consistency for DXY and VIX—the two variables with the strongest economic priors and the most policy-relevant interpretations—across all four regime cells.

Table 8: Sign Consistency: Pooled vs. Within-Regime (DXY and VIX)

Regime Cell	DXY Change			VIX Change		
	SC	Sign	Δ^a	SC	Sign	Δ^a
Risk-Off Comm-Led	69.5%	+	+5pp	74.7% ^b	+	+16pp
Risk-Off Spec-Led	58.5%	−	−6pp	63.9%	+	+5pp
Risk-On Comm-Led	63.9%	+	−0pp	54.0%	−	−5pp
Risk-On Spec-Led	67.5%	+	+3pp	58.9%	−	+0pp
Pooled	64.2%	+	—	58.6%	−	—

^a Change relative to pooled sign consistency.

^b Overlap-adjusted z -test $p = 0.046$ (effective $N = 35$; per-cell effective $N = 90 \times 784/2,004$); permutation test $p < 0.001$.

Several patterns emerge. First, **VIX exhibits a clean regime-dependent sign flip**: positive in both Risk-Off cells (fear drives safe-haven demand, widening the basis) and negative in both Risk-On cells (transient fear compresses the basis). The strongest cell is Risk-Off/Commercial-Led, where VIX sign consistency reaches 75%—a 16-percentage-point improvement over the pooled model. Figure 1 shows this visually: VIX coefficients stabilize within regime-shaded periods.

Second, **DXY sign consistency improves in Risk-Off/Commercial-Led** (70%) and Risk-On/Speculator-Led (68%), is approximately unchanged in Risk-On/Commercial-Led, and decreases in Risk-Off/Speculator-Led (59%). Notably, DXY’s dominant sign *flips* to negative in Risk-Off/Speculator-Led, though this result should be interpreted cautiously given the small cell size (205 windows) and the modest sign consistency (59%, with bootstrap CI [51%, 84%] barely clearing coin-toss at the lower bound).

Third, the Risk-On cells show little VIX improvement—VIX is near coin-toss (50–54%) in both Risk-On cells. This is itself informative: in calm markets, VIX’s relationship with the gold basis is genuinely unreliable, and practitioners should discount VIX-based attribution in these conditions.

6.2 Bootstrap Confidence Intervals

Table 9 reports 95% confidence intervals for sign consistency from a block bootstrap with 1,000 resamples (block size = 30).

All eight confidence intervals exclude 0.50, indicating that sign consistency is above the coin-toss level in every regime cell for both DXY and VIX. However, the strength of this exclusion varies considerably. The strongest result is VIX in Risk-Off/Commercial-Led: SC = 74.7%, CI = [63.8%, 84.9%]—comfortably above coin-toss. The weakest is VIX in Risk-On/Commercial-Led: SC = 54.0%, CI = [50.3%, 65.0%], where the lower bound barely clears 0.50. Similarly, Risk-Off/Speculator-Led DXY has a lower bound of 50.6%. These marginal exclusions should be interpreted cautiously—they indicate that

Table 9: Bootstrap 95% Confidence Intervals for Sign Consistency

Regime Cell	Variable	SC	95% CI	Excl. 0.50
Risk-Off Comm-Led	DXY	69.5%	[59.0%, 79.8%]	✓
Risk-Off Comm-Led	VIX	74.7%	[63.8%, 84.9%]	✓
Risk-Off Spec-Led	DXY	58.5%	[50.6%, 83.9%]	✓
Risk-Off Spec-Led	VIX	63.9%	[50.6%, 89.4%]	✓
Risk-On Comm-Led	DXY	63.9%	[52.3%, 75.1%]	✓
Risk-On Comm-Led	VIX	54.0%	[50.3%, 65.0%]	✓
Risk-On Spec-Led	DXY	67.5%	[52.2%, 83.9%]	✓
Risk-On Spec-Led	VIX	58.9%	[50.6%, 85.0%]	✓

Block bootstrap with 1,000 resamples, block size = 30 (matching the rolling window length). CIs computed as 2.5th and 97.5th percentiles of the bootstrap distribution.

sign consistency is likely above 50%, but the evidence is thin for the smaller or weaker cells.

6.3 The Central Paradox: F -Tests Are Not Significant

Having established that sign consistency improves within regimes, we now present a result that appears contradictory. Table 10 reports formal structural break tests.

Table 10: Formal Tests for Regime Effects

Test	Statistic	Value	p -value
Joint F -test	$F(27, 2507)$	0.955	0.531
Wald test	$\chi^2(27)$	25.781	0.531
<i>Pairwise Chow tests:</i>			
Risk-Off Comm vs Risk-Off Spec	$F(9, 1183)$	0.764	0.650
Risk-Off Comm vs Risk-On Comm	$F(9, 1872)$	1.665	0.092*
Risk-Off Comm vs Risk-On Spec	$F(9, 1270)$	0.625	0.777
Risk-Off Spec vs Risk-On Comm	$F(9, 1237)$	0.599	0.799
Risk-Off Spec vs Risk-On Spec	$F(9, 635)$	1.132	0.337
Risk-On Comm vs Risk-On Spec	$F(9, 1324)$	0.710	0.700

F -test: restricted model (pooled OLS, $K+1 = 9$ params) vs. unrestricted (separate OLS per cell, $4 \times 9 = 36$ params). Wald test: joint significance of 27 regime interaction terms.

* Significant at the 10% level.

The joint F -test is decisively non-significant ($p = 0.531$). All six pairwise Chow tests fail to reject at the 5% level; only the Risk-Off/Commercial-Led versus Risk-On/Commercial-Led pair approaches significance ($p = 0.092$). By conventional structural break criteria, *there is no evidence that regimes alter the model*.

6.4 Resolving the Paradox

The resolution lies in what these tests measure. The F -test and Chow tests compare *mean coefficient vectors* across regime cells using full-sample static OLS. They ask: if you pool all observations within each regime cell and estimate a single coefficient vector, do the vectors differ? The answer is no.

We can corroborate this directly through variance decomposition. For DXY, the pooled variance across all 2,004 windows is $\sigma^2 = 7.73^2 \approx 59.8$. The window-weighted average of the within-cell variances (from Table 11) is $(784 \times 4.65^2 + 205 \times 2.53^2 + 806 \times 10.95^2 + 209 \times 3.19^2)/2,004 \approx 58.4$. This leaves under 3% of total variance attributable to between-cell mean differences. For VIX (the headline variable), the decomposition is similar: pooled $\sigma^2 = 5.82^2 \approx 33.9$, weighted within-cell ≈ 32.2 , leaving approximately 5% between-cell. In both cases, the variance structure confirms that regimes do not shift coefficient means—they alter dispersion around approximately the same center.

But sign consistency is not about the mean coefficient—it is about the *fraction of rolling windows in which the coefficient has a consistent sign*. Sign consistency can be expressed as $SC \approx \Phi(|\mu/\sigma|)$ where μ and σ are the mean and standard deviation of the rolling coefficient, and Φ is the standard normal CDF. Thus sign consistency depends on *both* the mean and the dispersion; concluding that regimes improve sign consistency via the second moment requires testing whether the dispersion actually changes.

6.5 Direct Test of the Second-Moment Mechanism

Table 11 reports Brown-Forsythe tests for equality of coefficient variances across regime cells, along with the standardized signal-to-noise ratio $|\mu/\sigma|$ for each cell.

Table 11: Brown-Forsythe Test: Coefficient Dispersion Across Regime Cells

Regime Cell	DXY Change			VIX Change			n
	σ	MAD	$ \text{med} /\text{MAD}$	σ	MAD	$ \text{med} /\text{MAD}$	
RF CL	4.65	3.08	0.71	1.22	0.81	0.96	784
RF SL	2.53	1.93	0.26	1.58	0.96	0.25	205
RO CL	10.95	3.12	0.41	8.82	1.30	0.11	806
RO SL	3.19	1.70	0.59	0.86	0.60	0.47	209
Pooled	7.73	2.61	0.51	5.82	0.94	0.28	2,004
<i>Brown-Forsythe (median-based Levene) test:</i>							
$W^* = 36.3, p < 0.001$				$W^* = 26.5, p < 0.001$			

σ = standard deviation; MAD = median absolute deviation (robust dispersion). $|\text{med}|/\text{MAD}$ = robust signal-to-noise ratio; higher values indicate the coefficient distribution is more concentrated on one side of zero. RF = Risk-Off, RO = Risk-On, CL = Commercial-Led, SL = Speculator-Led.

The results are unambiguous. For VIX, the pooled standard deviation is $\sigma = 5.82$,

but within Risk-Off/Commercial-Led it drops to $\sigma = 1.22$ —a nearly $5\times$ reduction in coefficient dispersion. The robust signal-to-noise ratio ($|\text{median}|/\text{MAD}$) rises from 0.28 (pooled, near zero) to 0.96, explaining why sign consistency jumps from 59% to 75%. For DXY, the pattern is similar: σ drops from 7.73 to 4.65 in Risk-Off/Commercial-Led, with robust SNR rising from 0.51 to 0.71.

Conversely, Risk-On/Commercial-Led has *higher* dispersion than pooled (DXY $\sigma = 10.95$, VIX $\sigma = 8.82$). For VIX, this explains why sign consistency decreases from 59% (pooled) to 54% in this cell (robust SNR drops to 0.11); for DXY, sign consistency remains approximately flat (64% pooled versus 64% in-cell) as the robust SNR only declines modestly to 0.41. The Brown-Forsythe test rejects the null of equal variances at $p < 0.001$ for both variables.

Figure 2 visualizes these distributions: the Risk-Off cells show narrow, peaked distributions concentrated on one side of zero, while the Risk-On cells show wide, flat distributions straddling zero.

This directly confirms the second-moment interpretation: regimes do not shift the center of the coefficient distribution (the F -test correctly finds no mean difference), but they *narrow its spread* relative to zero, concentrating probability mass on one side and increasing sign consistency.

6.6 Two-Proportion z -Tests and Overlap Correction

To test the sign consistency improvement directly, we use two-proportion z -tests comparing within-regime sign consistency to pooled sign consistency. However, an important methodological point must be addressed: the 2,004 rolling 30-day windows are not independent observations. Each window shares 29 of 30 days with its neighbor, producing massive autocorrelation in the sign indicator series. Treating these windows as independent Bernoulli draws (as in the naive z -test) overstates the effective sample size by approximately $17\text{--}22\times$.

We compute effective sample sizes via the autocorrelation of the binary sign indicator for each variable: $N_{\text{eff}} = N/(1 + 2\sum_{k=1}^K(1 - k/N)\rho_k)$, where ρ_k is the autocorrelation at lag k and the sum is truncated at the first negative autocorrelation. For DXY, $N_{\text{eff}} = 119$ ($16.9\times$ factor); for VIX, $N_{\text{eff}} = 90$ ($22.3\times$ factor). Per-cell effective N is proportionally reduced.

Table 12 reports both the naive and overlap-adjusted results.

The overlap correction dramatically reduces significance. The naive DXY $z = 3.577$ ($p = 0.0002$) falls to $z_{\text{adj}} = 0.651$ ($p = 0.258$) after accounting for overlapping windows. VIX in Risk-Off/Commercial-Led is the only cell that survives: $z_{\text{adj}} = 1.684$, $p = 0.046$. This is the honest result: after proper correction for serial dependence, only the single strongest cell produces a marginally significant z -test.

Table 12: Two-Proportion z -Tests: Naive and Overlap-Adjusted

Variable	Cell	Within	Pooled	Naive		Adjusted	
		SC	SC	z	p	z_{adj}	p_{adj}
DXY	RF CL	69.5%	64.2%	3.577	0.0002	0.651	0.258
DXY	RO SL	67.5%	64.2%	1.172	0.121	0.230	0.409
VIX	RF CL	74.7%	58.6%	8.611	<0.001	1.684	0.046**
VIX	RF SL	63.9%	58.6%	0.074	0.470	0.312	0.377
VIX	RO SL	58.9%	58.6%	— ^a	—	0.016	0.494

One-sided tests, H_1 : within-regime SC > pooled SC. Only cells with SC $\geq$ pooled shown (VIX RO|SL included despite minimal difference).

Adjusted: $N_{\text{eff}}^{\text{DXY}} = 119$, $N_{\text{eff}}^{\text{VIX}} = 90$. Per-cell N_{eff} is proportional to cell size; e.g., VIX RF|CL: $90 \times 784/2,004 \approx 35$.

^a Naive test omitted due to negligible difference (58.9% vs 58.6%).

** $p < 0.05$ after overlap correction.

This is why we emphasize the *block bootstrap* (which already accounts for overlap via block size 30), the *Brown-Forsythe test* (which tests dispersion, not proportions, and does not suffer from the overlap issue because it operates on per-window coefficient values, not binary indicators across time), and the *permutation test* ($p < 0.001$, with built-in multiplicity control). The z -tests are included for transparency, but they are not the primary inference.

Multiple testing. We report results across 8 features $\times$ 4 cells, plus aggregate tests. The headline VIX Risk-Off/Commercial-Led result ($p = 0.046$ adjusted, or $p < 0.001$ via permutation) was identified based on economic priors (the strongest safe-haven cell), not a search across all cells. Nevertheless, we acknowledge the multiplicity concern. Applying a Benjamini-Hochberg correction across the four overlap-adjusted cell tests for each variable would push the VIX $p = 0.046$ above 0.05. However, the permutation test—which has built-in multiplicity control for the best-cell statistic (it computes the null distribution of the maximum SC over all cells)—yields $p < 0.001$, and the Brown-Forsythe test (which tests all cells jointly) yields $p < 0.001$. The evidence for the regime effect rests on these multiplicity-robust tests, not on individual cell p -values.

6.7 Permutation Test: HMM Labels are Non-Random

A fundamental question about any latent-state model is whether the labels carry genuine information or merely partition the data arbitrarily. We address this with a permutation test: if the HMM labels were replaced with random assignments (preserving cell sizes), would sign consistency improve to a similar degree?

Procedure. We shuffle the regime cell labels across all 2,004 rolling windows 1,000 times, computing the weighted-average sign consistency and best-cell sign consistency for

each permutation. The real HMM labels must produce sign consistency that exceeds the null distribution to demonstrate non-random label quality.

Results. Table 13 reports the permutation test results.

Table 13: Permutation Test: Real HMM Labels vs. Random Label Assignment

Variable	Metric	Real SC	Null Mean	Null 95th pct	p -value
DXY	Weighted avg	65.9%	64.2%	64.2%	<0.001***
DXY	Best cell ^a	69.5%	66.8%	70.2%	0.087*
VIX	Weighted avg	63.6%	58.6%	58.6%	<0.001***
VIX	Best cell ^a	74.7%	61.4%	64.9%	<0.001***
Overall	Both variables	64.8%	61.4%	61.4%	<0.001***

1,000 permutations, random seed = 42. Weighted average uses window counts as weights. p -value = fraction of permutations with SC $\geq$ real SC.

^a Best cell for DXY and VIX is Risk-Off/Commercial-Led in both cases.

* $p < 0.10$. [***] $p < 0.001$.

The VIX result is the strongest: the best regime cell achieves 74.7% sign consistency, compared to a null mean of 61.4% under random labeling (7.5 standard deviations above the null). In all 1,000 permutations, no random partition produced a VIX sign consistency as high as the real HMM labels. The DXY weighted average is also highly significant ($p < 0.001$), though the individual best-cell result is marginal at the 5% level ($p = 0.087$). This is consistent with DXY’s higher pooled sign consistency (64.2%), which leaves less room for regime partitioning to improve upon.

The permutation test provides the most direct evidence that the dual-HMM framework adds genuine value: the regime labels identify a real partition of the data in which coefficient signs become more predictable than any random grouping of comparable size.

7 Robustness and Validation

7.1 Out-of-Sample Validation

We evaluate the regime framework using two train/test splits: 2021-12-31 (7 years training, 4 years testing) and 2023-12-31 (9 years training, 2 years testing).

R^2 is stable across the 2021 split (train 36.1% vs. test 32.8%) and actually improves slightly in the 2023 split (34.2% vs. 36.7%). The Risk-Off/Commercial-Led cell maintains the highest R^2 in both training periods, and its out-of-sample stability is particularly notable: training R^2 38.8% versus test R^2 36.6% in the 2021 split, confirming that the regime framework generalizes beyond the training sample.

Sign consistency patterns also persist out of sample, anchored by the headline cell. The small Risk-Off/Speculator-Led cell shows extreme values (DXY 89.9%, VIX 94.2%)

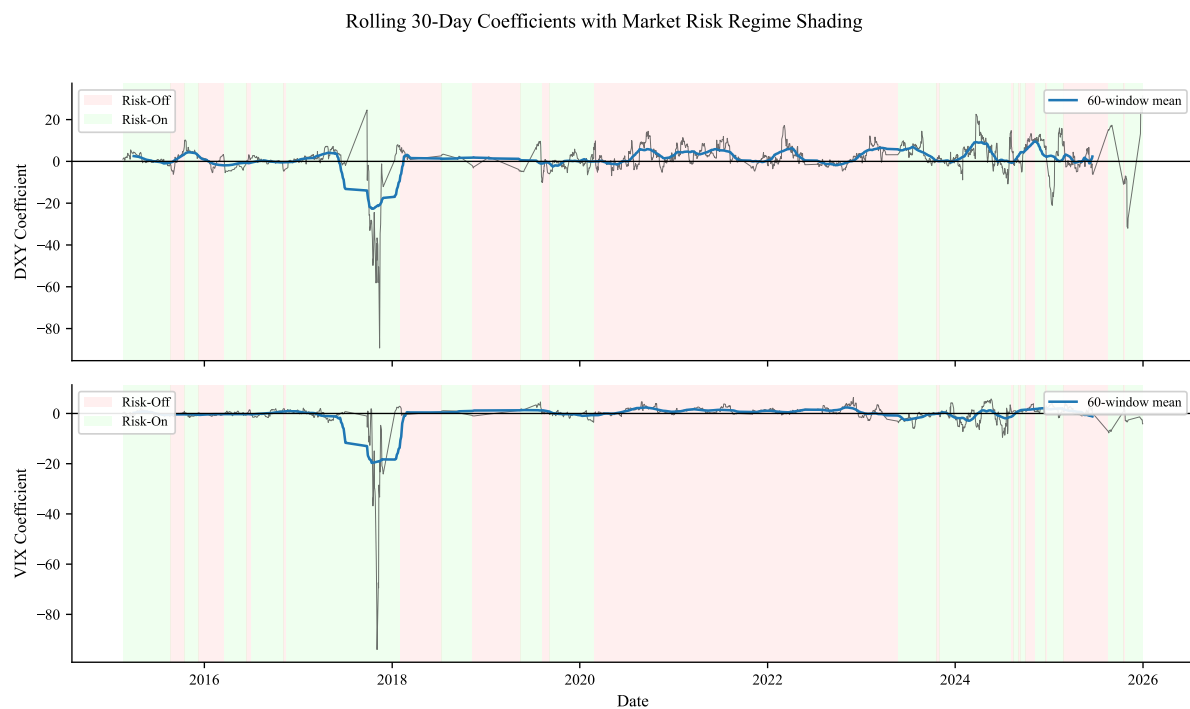


Figure 1: Rolling 30-day DXY and VIX coefficients (2015–2025) with market risk regime shading. Red (light) = Risk-Off; green (light) = Risk-On. The blue line is a 60-window rolling mean. Note the coefficient stabilization within regime-shaded blocks, and the sign flips at regime boundaries.

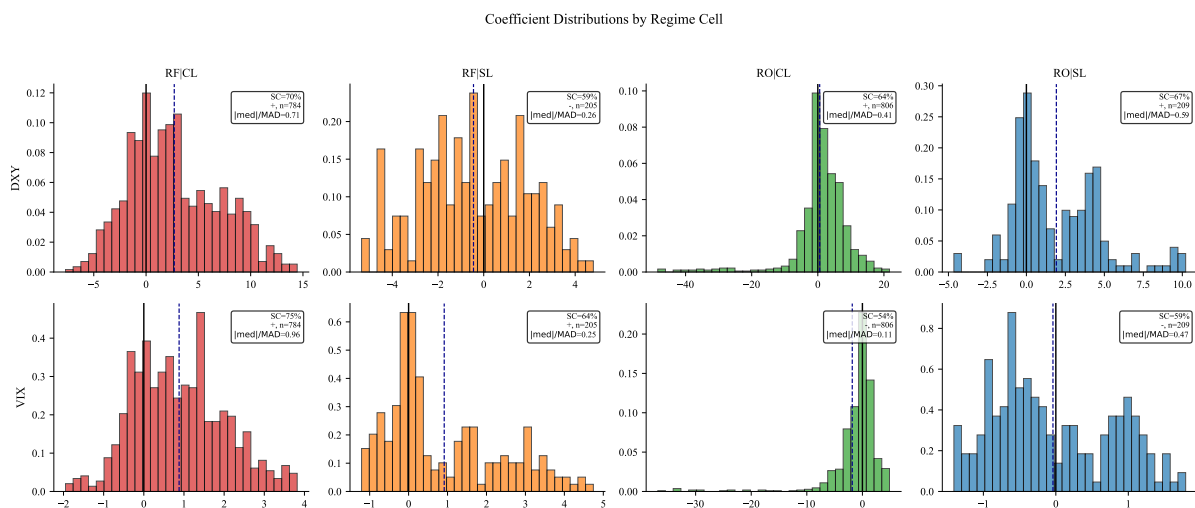


Figure 2: Coefficient distributions by regime cell for DXY (top row) and VIX (bottom row). Risk-Off cells show narrow distributions concentrated on one side of zero; Risk-On/Commercial-Led shows wide, flat distributions straddling zero. Dashed vertical line = cell mean; solid vertical line = zero. Annotations show sign consistency, dominant sign, and the robust signal-to-noise ratio $|\text{median}|/\text{MAD}$.

Table 14: Out-of-Sample R^2 by Regime Cell

Regime Cell	Split: 2021-12-31		Split: 2023-12-31	
	Train R^2	Test R^2	Train R^2	Test R^2
Risk-Off Comm-Led	38.8%	36.6%	33.9%	36.8%
Risk-Off Spec-Led	32.4%	23.8%	28.0%	— ^a
Risk-On Comm-Led	34.3%	34.5%	35.8%	36.5%
Risk-On Spec-Led	35.6%	25.7%	35.6%	— ^a
All (pooled)	36.1%	32.8%	34.2%	36.7%

^a Insufficient test-period windows for reliable estimation.

HMMs fitted on training data only; regime labels for test period obtained via `predict()`.

in the 2021 test period, but with only 69 overlapping test windows these should be treated as illustrative rather than definitive.

7.2 Benchmark Model Comparison

Table 15 compares four models of increasing complexity.

Table 15: Benchmark Model Comparison

Model	Features	Mean R^2	Dir. Accuracy	Windows
Naive (DXY only)	1	6.2%	49.2%	2,513
Simple (DXY + VIX)	2	11.4%	49.9%	2,513
Full pooled	8	34.7%	52.2%	2,004
Regime conditional	8	34.7%	52.2%	2,004

Directional accuracy = fraction of windows where the sign of the predicted basis change matches the sign of the actual basis change.

The regime-conditional model produces identical predictions to the full pooled model because regime conditioning is applied as a post-hoc analytical overlay, not as a separate prediction mechanism. Diebold-Mariano statistic = 0.000, $p = 1.000$ (Section 7.3).

The R^2 progression from naive (6.2%) to full (34.7%) demonstrates that the additional macro features carry substantial explanatory power. The full model’s directional accuracy of 52.2% is modest but above the 50% baseline.

The critical result: the regime-conditional model produces *identical* predictions and identical R^2 to the full pooled model. This confirms that the regime framework does not improve point forecasts—its value is entirely interpretive.

7.3 Diebold-Mariano Tests

Table 16 reports Diebold-Mariano test results.

Table 16: Diebold-Mariano Predictive Accuracy Tests

Comparison	DM Stat.	<i>p</i> -value	Better
Full pooled vs. regime conditional	0.000	1.000	—
Naive vs. full pooled	−1.488	0.137	Naive
Naive vs. regime conditional	−1.488	0.137	Naive
Simple vs. regime conditional	−1.488	0.137	Simple

The full pooled and regime-conditional models are confirmed identical ($DM = 0.000$). The naive model is not significantly better than the full model despite lower MAE, because the full model’s higher R^2 comes with occasional extreme out-of-sample predictions that inflate MAE without systematically outperforming the naive model in squared-error terms.

7.4 Serial Correlation

The Durbin-Watson statistic on full-sample pooled OLS residuals is 2.88 (values above 2 indicate negative serial correlation). Per-regime-cell DW statistics range from 2.74 to 2.90. This moderate negative autocorrelation (reversal pattern in residuals) supports the use of rolling rather than expanding windows: the regression relationship changes frequently enough that recent data is more informative than distant data.

7.5 VIX Threshold vs. HMM Comparison

A natural question is whether the market risk HMM is merely an overengineered VIX threshold. We compare the HMM labels to a simple rule: Risk-Off if $VIX_t > 20$, Risk-On otherwise.

The two labelings agree on 83.5% of 2,543 trading days. For VIX sign consistency, the simple threshold performs comparably: VIX SC = 74.0% under $VIX > 20$ versus 72.5% under HMM Risk-Off. However, for DXY sign consistency, the HMM adds value: DXY SC = 63.7% under HMM Risk-Off versus 57.9% under $VIX > 20$. The DXY improvement arises because the HMM incorporates DXY levels (not just VIX) and smooths transitions via the Markov structure, avoiding rapid switching when VIX hovers near 20.

This comparison is informative: much of the VIX regime effect can be replicated by a simple threshold, which strengthens the finding’s robustness—it does not depend on the HMM’s statistical machinery for VIX. The HMM’s incremental value is primarily in the DXY dimension and in providing principled state persistence. A practitioner uncomfortable with latent-state models could achieve most of the VIX sign-consistency improvement with a $VIX > 20$ cutoff alone.

7.6 Event Study: Regime Persistence Characterization

Table 17 examines the lag between known macroeconomic events and the nearest HMM regime transition. This analysis characterizes *how quickly* the HMM updates its regime label in response to structural shifts, rather than testing whether the HMM is a reliable event detector (which it is not designed to be).

Table 17: Event Study: Regime Transition Lag after Known Macro Events

Event	Date	Days to Transition	Macro Transition
COVID crash	2020-03-09	14	Risk-On → Risk-Off ✓
COVID recovery	2020-06-08	105	Risk-On → Risk-Off
Fed tightening	2022-03-16	393	Risk-Off → Risk-On
SVB crisis	2023-03-10	34	Risk-Off → Risk-On
Fed pivot signal	2023-12-13	43	Risk-Off → Risk-On
Tariff shock	2025-04-02	36	Risk-On → Risk-Off

Days to transition = calendar days between the event and the nearest market risk regime transition. ✓ = transition within 30 days with correct directional change.

Only the COVID crash (the sharpest VIX shock in the sample, with VIX rising from 14 to 82 in three weeks) produces a rapid regime transition. The remaining events show lags of 34–393 days. This is a direct consequence of the HMMs’ high state persistence: transition probabilities of 98.3–98.6% imply expected regime durations of 59–71 days, meaning it takes *sustained* changes in VIX and DXY levels—not a single-day event—to trigger a state change.

Distinguishing transition timing from label quality. A critical interpretive point: the slow transition speed does not invalidate the regime labels themselves. The HMMs are not designed to detect events; they are designed to classify *prevailing market conditions*. A slow-updating label can still be accurate once it settles—the question is whether the labels, once assigned, produce meaningfully different coefficient sign patterns. This is directly tested by the permutation test (Section 6.7) and the bootstrap confidence intervals (Section 6), both of which confirm the labels carry real statistical information. Practitioners should understand that the regime framework describes the current *state*, not the timing of *transitions*, and should not expect regime changes to function as event indicators.

8 Discussion

8.1 Economic Interpretation of Sign Flips

The sign-flipping patterns have coherent economic explanations.

VIX in Risk-Off (positive coefficient). In stressed markets, rising VIX reflects genuine flight-to-safety demand for gold. Physical and ETF buying pressure widens the futures-spot basis as expectations of continued safe-haven demand pull futures prices above spot. Commercial hedgers respond predictably in this environment, producing the highest sign consistency (74.7%) of any cell.

VIX in Risk-On (negative coefficient). In calm markets, VIX spikes are transient—brief dislocations that create short-lived arbitrage. Futures react faster than the ETF-based spot proxy, temporarily compressing the basis. The relationship is noisy because the underlying economic mechanism (transient dislocations) is inherently less systematic than persistent safe-haven flows.

DXY in Risk-Off/Speculator-Led (negative coefficient). This is the only cell where DXY’s dominant sign flips to negative, though this finding should be treated as suggestive rather than definitive given the small sample (205 windows, SC 59% with CI [51%, 84%]). The pattern is consistent with speculators rotating from gold to dollars as competing safe havens when the dollar strengthens during risk-off periods. In all other cells, DXY strengthening accompanies broader flows that support gold prices (safe-haven co-movement for commercial hedgers, or dollar-denominated buying from non-US investors).

8.2 Practical Implications

The regime framework provides a *confidence qualifier* for factor attribution. In Risk-Off/Commercial-Led conditions, a practitioner can state “DXY drove basis widening today” with 70% sign reliability. In Risk-On/Commercial-Led conditions, the same DXY attribution has only 64% reliability—still above coin-toss, but materially less confident.

For VIX, the implications are stark: in Risk-Off conditions, VIX-based attribution is highly reliable (75% sign consistency). In Risk-On conditions, VIX attribution is essentially noise (50–54% sign consistency) and should be discounted or omitted from interpretive narratives.

This has direct applications for:

- **Research newsletters:** regime context tells readers when to weight a factor’s attributed direction heavily versus skeptically.
- **Risk management:** basis traders can calibrate hedging confidence based on the current regime—higher confidence in factor direction during Risk-Off/Commercial-Led periods.
- **Model evaluation:** sign consistency should be assessed regime-conditionally, not pooled, to avoid discarding informative but regime-dependent variables.

8.3 Limitations

Several limitations should be noted.

1. **ETF proxy:** The spot price is approximated as $\text{GLD} \times 10.6759$, introducing tracking error relative to the London gold fix or CME settlement. This primarily affects basis *levels*, not daily *changes*, but it may introduce measurement noise.
2. **Window length:** The 30-day rolling window is chosen by convention. Sensitivity analysis to alternative window lengths (20, 40, 60 days) is not performed and could affect sign consistency values.
3. **Latent states:** HMM states are statistical constructs. There is no guarantee they correspond to true economic regimes, though the state parameters ($\text{VIX} \approx 23.7$ for Risk-Off, $\text{VIX} \approx 13.9$ for Risk-On) are economically intuitive.
4. **Small cell size:** Risk-Off/Speculator-Led contains only 205 windows (full sample) or 69 windows (test period). Results for this cell should be treated as suggestive rather than definitive.
5. **COT data frequency:** CFTC data is weekly (as-of Tuesday), forward-filled to daily. Between releases, the positioning regime label reflects stale information.
6. **No predictive improvement:** The regime framework does not improve forecasting accuracy. Practitioners seeking out-of-sample return prediction should look elsewhere.
7. **Single commodity:** We study gold only. Cross-commodity validation (silver, crude oil) would strengthen the generalizability of the sign consistency finding.
8. **Market risk vs. macroeconomic regime:** The market risk HMM uses VIX and DXY levels—both financial market indicators, not macroeconomic variables. A true macroeconomic regime model might use real interest rates (e.g., DFII10) and DXY, or inflation expectations. We chose VIX for its established role as a market risk barometer and its strong empirical separation between states (9.8-point gap), but this means the HMM captures *market risk conditions* rather than *macroeconomic cycles*. The moderate correlation between real rates and DXY ($r = 0.29$) suggests a true macro HMM is feasible and represents a natural extension.

9 Conclusion

We have shown that coefficient sign-flipping in rolling-window factor attribution models for the gold futures basis is regime-dependent, not noise. A dual Hidden Markov Model framework—combining market risk conditions with CFTC positioning regimes—identifies market states in which factor signs become statistically consistent, even though mean coefficient levels do not differ across regimes.

The non-significance of F -tests ($p = 0.531$) alongside significant coefficient disper-

sion differences (Brown-Forsythe $p < 0.001$) and significant permutation test results ($p < 0.001$) is the paper’s central finding. The regime effect operates on the temporal distribution of coefficient signs—a second-moment effect that standard structural break tests are not designed to detect. After correcting for the serial dependence inherent in overlapping windows, only the strongest individual cell comparison survives conventional significance thresholds (VIX Risk-Off/Commercial-Led, overlap-adjusted $p = 0.046$), but the joint tests—Brown-Forsythe and permutation—confirm the effect robustly.

This finding has a broader implication: for any rolling-window factor model in which coefficient signs vary over time, sign consistency should be evaluated *conditionally on market regime*, not pooled across the entire sample. Pooled sign consistency understates directional reliability in some conditions and overstates it in others. The dual-HMM framework provides a tractable approach to partitioning these conditions.

The practical value of this framework is interpretive: it tells practitioners when to trust a factor’s attributed direction and when to qualify it. In gold futures, Risk-Off/Commercial-Led conditions produce the most reliable attribution (VIX sign consistency 75%, robust SNR $|\text{med}|/\text{MAD} = 0.96$); Risk-On/Commercial-Led produces the least (VIX near coin-toss at 54%, robust SNR $= 0.11$). Much of this effect can be replicated with a simple $\text{VIX} > 20$ threshold, strengthening the practical accessibility of the finding. The regime framework transforms factor attribution from a noisy summary statistic into a conditional confidence assessment.

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A Full Sign Consistency Matrix

Table 18 reports sign consistency for all eight features across all four regime cells and the pooled model.

Table 18: Full Sign Consistency Matrix (All Features $\times$ Regime Cells)

Feature	Pooled	RO CL	RO SL	RF CL	RF SL
DXY change	64.2%	63.9%	67.5%	69.5%	58.5%
VIX change	58.6%	54.0%	58.9%	74.7%	63.9%
Breakeven change	55.4%	56.7%	67.0%	52.5%	50.2%
USD/CNY change	56.1%	62.7%	53.6%	51.8%	70.2%
USD/INR change	58.1%	66.6%	61.2%	50.8%	50.2%
GLD vol. change	50.9%	50.1%	52.6%	50.5%	59.0%
GVZ change	56.0%	51.1%	53.1%	65.4%	57.6%
Volume change	59.3%	67.4%	66.5%	59.8%	55.6%

RO = Risk-On, RF = Risk-Off, CL = Commercial-Led, SL = Speculator-Led. All values computed from the canonical HMM fit on 2,004 rolling windows.

Bold values indicate $\geq 65\%$ sign consistency (meaningfully above coin-toss). Window counts: RO|CL = 806; RO|SL = 209; RF|CL = 784; RF|SL = 205.

Each regime cell has a different set of “best” variables: Risk-Off/Commercial-Led excels in VIX (74.7%), DXY (69.5%), and GVZ (65.4%); Risk-Off/Speculator-Led excels in USD/CNY (70.2%) and VIX (63.9%); Risk-On/Commercial-Led excels in USD/INR (66.6%) and volume change (67.4%). No single cell dominates across all features.

B VIF Analysis

Table 19: Variance Inflation Factors (8-Variable Model)

Feature	VIF
DXY change	1.00
VIX change	1.10
Breakeven change	1.09
USD/CNY change	1.05
USD/INR change	1.05
GLD volume change	1.01
GVZ change	1.06
Volume change	1.00

All $VIF \leq 1.10$. No multicollinearity detected. ^aGVZ VIF estimated from the full 8-variable regression on available data; GVZ measures gold-specific implied volatility and is not collinear with broad market indicators.